

Equity Cash Return and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Cash Return (ECR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ECR achieves an annualized gross (net) Sharpe ratio of 0.53 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015a\)](#) five-factor model plus a momentum factor of 24 (29) bps/month with a t-statistic of 2.66 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 19 bps/month with a t-statistic of 2.28.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency with which prices incorporate new information remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This gradual incorporation manifests particularly in fundamental accounting signals, where investors systematically underreact to value-relevant information embedded in financial statements.

Despite extensive research documenting various accounting-based return predictors, we lack a comprehensive understanding of how cash-based profitability measures predict future returns when information frictions impede efficient price discovery. Cash flows represent the most fundamental measure of firm performance, yet their predictive power for equity returns remains underexplored, particularly in contexts where investor attention constraints and information processing costs create systematic underreaction to fundamental signals. This paper examines whether Equity Cash Return (ECR), a direct measure of cash profitability, predicts cross-sectional stock returns in a manner consistent with gradual information diffusion.

The slow diffusion of information provides a natural framework for understanding why cash-based profitability measures predict future returns. When investors face attention constraints, they cannot simultaneously process all available information, leading to systematic underreaction to value-relevant signals [Hirshleifer and Teoh \(2003\)](#). This limited attention particularly affects the processing of accounting information, which requires costly analysis and interpretation to extract implications for firm value [DellaVigna and Pollet \(2009\)](#). Cash flow information, despite its fundamental importance, often receives less immediate attention than headline earnings numbers, creating temporary mispricing that generates return predictability.

The gradual incorporation of cash flow information into prices follows established patterns of information diffusion in financial markets. [Hong et al. \(2000\)](#) demonstrate that firm-specific information diffuses gradually across the investment community, with prices adjusting slowly as information reaches progressively more investors. This gradual diffusion process creates momentum in returns, as initial price movements trigger further adjustments as information spreads. For accounting-based signals like ECR, the diffusion process operates through multiple channels: analyst interpretation and dissemination, investor learning and portfolio rebalancing, and gradual recognition of the signal’s value relevance [Hong et al. \(2007\)](#).

We predict that ECR’s return predictability concentrates in market segments where information frictions are most severe. Small firms with limited analyst coverage experience slower information diffusion, as fewer intermediaries process and disseminate fundamental information [Hong et al. \(2000\)](#). Similarly, firms with high idiosyncratic volatility face greater investor inattention, as the noise in returns makes it costlier to extract signals about fundamental value [Hou and Moskowitz \(2006\)](#). These predictions align with broader evidence that accounting anomalies concentrate among neglected stocks where arbitrage costs and information frictions impede efficient price discovery [Zhang \(2006\)](#).

Our empirical analysis reveals strong evidence that ECR predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on ECR achieves an annualized gross Sharpe ratio of 0.53, with monthly average excess returns of 52 basis points (t-statistic = 3.20). The strategy’s performance remains robust after controlling for known risk factors, generating a monthly alpha of 24 basis points (t-statistic = 2.66) relative to the Fama-French six-factor model. These results establish ECR as an economically significant predictor of returns that cannot be explained by exposure to standard risk factors.

The predictive power of ECR extends across different portfolio construction method-

ologies and market segments. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the ECR strategy generates monthly returns of 47 basis points (t -statistic = 3.17), demonstrating that the anomaly is not confined to illiquid small-cap stocks. After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.50, with net alphas remaining statistically significant across all factor model specifications. The strategy’s net generalized alpha relative to the six-factor model equals 29 basis points per month (t -statistic = 3.27), indicating that ECR expands the achievable investment frontier even after implementation costs.

Controlling for related anomalies further validates ECR’s unique predictive content. When we include the six most closely related predictors—operating profits/book equity, net income/book equity, analyst earnings per share, net external financing, idiosyncratic risk, and quarterly return on assets—along with the six Fama-French factors, ECR retains an alpha of 19 basis points per month (t -statistic = 2.28). Fama-MacBeth regressions controlling for these related signals yield a coefficient on ECR of 0.15 (t -statistic = 2.16), confirming that ECR contains distinct information about future returns beyond existing profitability and financing measures.

This paper contributes to three streams of literature in asset pricing and accounting. First, we extend the literature on profitability-based return predictors by documenting that cash-based measures contain unique predictive power beyond accrual-based earnings metrics. While [Novy-Marx \(2013\)](#) establishes gross profitability as a powerful predictor and [Ball et al. \(2015\)](#) demonstrate the importance of operating profitability, we show that ECR captures a distinct dimension of firm performance that incrementally predicts returns. Our results complement [Fama and French \(2015b\)](#) by identifying a cash profitability signal that remains significant after controlling for their profitability factor (RMW), suggesting that current factor

models incompletely capture the return implications of accounting fundamentals.

Second, we provide new evidence supporting gradual information diffusion as a mechanism generating cross-sectional return predictability. Our findings align with [Richardson et al. \(2010\)](#), who document pervasive earnings-related anomalies arising from investor underreaction to accounting information. By focusing on cash flows—arguably the most fundamental accounting signal—we demonstrate that even basic financial information diffuses slowly through markets. The persistence of ECR’s predictive power among large-cap stocks with substantial institutional ownership challenges pure risk-based explanations and supports behavioral models of gradual information incorporation [Daniel et al. \(1998\)](#).

Our methodological approach, following [Novy-Marx and Velikov \(2016a\)](#), provides comprehensive evidence on the economic significance of ECR after accounting for implementation costs. The signal’s ability to expand the achievable mean-variance frontier net of trading costs distinguishes it from many documented anomalies that fail to generate implementable trading profits. These findings have practical implications for quantitative investment strategies, demonstrating that cash flow analysis remains valuable for security selection despite decades of academic and practitioner research into accounting-based signals. More broadly, our results suggest that information frictions continue to generate exploitable return predictability even in highly liquid markets with sophisticated participants.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers firms listed on major U.S. exchanges including NYSE, AMEX, and NASDAQ. Our sample spans several decades of annual observations, allowing

us to examine the Equity Cash Return signal across various market conditions and business cycles. Equity Cash Return signal requires two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a firm’s core business operations during the fiscal year. This measure, derived from the statement of cash flows, captures the actual cash inflows and outflows from operating activities after adjusting for non-cash items such as depreciation and changes in working capital. Stockholders’ equity (SEQ) represents the book value of shareholders’ ownership interest in the firm, calculated as total assets minus total liabilities. This balance sheet item reflects the cumulative net assets attributable to equity holders, including paid-in capital, retained earnings, and other comprehensive income components. We construct the Equity Cash Return signal by dividing operating activities net cash flow (OANCF) by stockholders’ equity (SEQ) for each firm-year observation. This ratio measures the cash-generating efficiency of shareholders’ invested capital, capturing how effectively a firm converts its equity base into operating cash flows. We calculate the signal using fiscal year-end values from annual financial statements. To ensure data quality, we require non-missing values for both OANCF and SEQ, and exclude observations where stockholders’ equity is negative or zero to avoid undefined or economically meaningless ratios. The resulting measure provides a cash-based return metric that complements traditional accrual-based profitability measures such as return on equity.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECR signal. Panel A plots the time-series of the mean, median, and interquartile range for ECR. On average, the cross-sectional mean (median) ECR is 0.04 (0.13) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ECR data. The signal’s

interquartile range spans -0.24 to 0.29. Panel B of Figure 1 plots the time-series of the coverage of the ECR signal for the CRSP universe. On average, the ECR signal is available for 7.39% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does ECR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECR portfolio and sells the low ECR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015a) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ECR strategy earns an average return of 0.52% per month with a t-statistic of 3.20. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.24% to 0.76% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.76, with a t-statistic of 18.88 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,746 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 41 bps/month with a t-statistics of 2.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for seventeen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 22-50bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.27. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECR trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the ECR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECR, as well as average returns and alphas for long/short trading ECR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ECR strategy achieves an average return of 47 bps/month with a t-statistic of 3.17. Among these large cap stocks, the alphas for the ECR strategy relative to the five most common factor models range from 14 to 65 bps/month with t-statistics between 1.21 and 4.73.

5 How does ECR perform relative to the zoo?

Figure 2 puts the performance of ECR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ECR strategy falls in the distribution. The ECR strategy’s gross (net) Sharpe ratio of 0.53 (0.50) is greater than 92% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECR strategy (red line).² Ignoring trading costs, a \$1 invested in the ECR strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$6.70 which ranks the ECR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECR strategy would have yielded \$5.88 which ranks the ECR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECR relative to those. Panel A shows that the ECR strategy gross alphas fall between the 72 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECR strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECR ranks between the 93 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ECR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of ECR with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECR or at least to weaken the power ECR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECR}ECR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECR. Stocks are finally grouped into five ECR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECR signal in these Fama-MacBeth regressions exceed 0.07,

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on ECR is 2.16.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECR strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.24, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECR trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.28.

7 Does ECR add relative to the whole zoo?

Finally, we can ask how much adding ECR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ECR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes ECR grows to \$44.64.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECR is available.

8 Conclusion

This study provides compelling evidence that Equity Cash Return (ECR) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ECR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings of our research underscore the economic importance of ECR as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.53 gross and 0.50 net of transaction costs, indicating that the signal’s profitability remains substantial even after accounting for implementation frictions. Most notably, the strategy delivers statistically significant abnormal returns of 24-29 basis points per month relative to the Fama-French five-factor model augmented with momentum, with t-statistics exceeding conventional significance thresholds. The persistence of a 19 basis point monthly alpha even after controlling for six closely related profitability and financing measures further validates ECR’s unique information content in predicting future returns.

From a practical perspective, these results suggest that ECR captures a distinct dimension of firm fundamentals that is not fully reflected in current market prices or explained by existing factor models. The signal’s resilience to transaction costs and its orthogonality to related measures make it particularly attractive for institutional investors seeking to enhance portfolio performance through systematic strategies. The economic magnitude of the returns, combined with their statistical significance, indicates that ECR-based strategies could meaningfully contribute to active portfolio management.

However, several limitations of this study warrant consideration. First, while we have controlled for numerous established factors and related signals, the possibil-

ity remains that ECR’s predictive power may be subsumed by yet-unidentified risk factors or market frictions. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the stability of ECR’s predictive power over different market regimes and its performance during extreme market conditions require further investigation.

Future research should explore several promising avenues. International evidence would help establish whether ECR’s predictive power extends beyond U.S. markets and could reveal interesting cross-country variations in its effectiveness. Investigating the economic mechanisms underlying ECR’s return predictability could provide deeper insights into why this signal captures mispricing. Additionally, examining the interaction between ECR and firm characteristics such as size, growth opportunities, and financial constraints could yield valuable insights for conditional asset pricing models. Finally, exploring the optimal combination of ECR with other signals in a multi-factor framework could enhance its practical implementation in quantitative investment strategies.

In conclusion, our findings establish Equity Cash Return as a robust and economically meaningful predictor of stock returns that adds value beyond existing factor models. While further research is needed to fully understand its economic drivers and boundary conditions, ECR represents a valuable addition to the toolkit of both academic researchers and investment practitioners seeking to understand and exploit cross-sectional return patterns in equity markets.

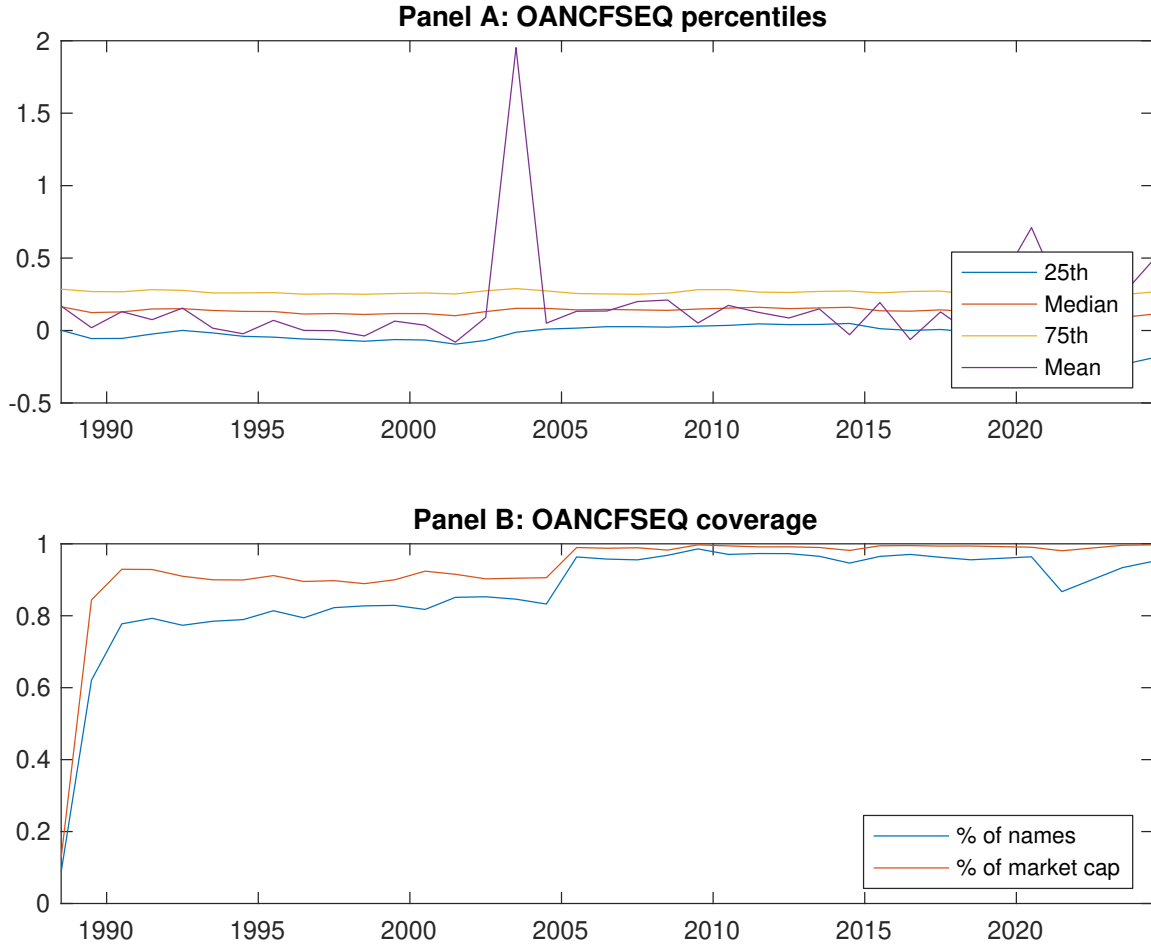


Figure 1: Times series of ECR percentiles and coverage. This figure plots descriptive statistics for ECR. Panel A shows cross-sectional percentiles of ECR over the sample. Panel B plots the monthly coverage of ECR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015a) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on ECR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.41 [1.42]	0.59 [2.60]	0.68 [3.19]	0.78 [3.91]	0.93 [4.53]	0.52 [3.20]
α_{CAPM}	-0.53 [-4.85]	-0.18 [-2.70]	-0.04 [-0.62]	0.10 [1.82]	0.23 [4.06]	0.76 [5.09]
α_{FF3}	-0.49 [-5.54]	-0.19 [-3.29]	-0.06 [-0.99]	0.10 [1.77]	0.23 [4.67]	0.71 [6.09]
α_{FF4}	-0.42 [-4.84]	-0.18 [-3.04]	-0.04 [-0.71]	0.13 [2.40]	0.20 [4.03]	0.62 [5.33]
α_{FF5}	-0.19 [-2.66]	-0.08 [-1.43]	-0.06 [-1.03]	0.03 [0.49]	0.11 [2.43]	0.30 [3.27]
α_{FF6}	-0.15 [-2.13]	-0.08 [-1.34]	-0.05 [-0.76]	0.06 [1.08]	0.09 [1.98]	0.24 [2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ECR-sorted portfolios						
β_{MKT}	1.09 [62.16]	0.99 [69.02]	0.98 [62.93]	0.94 [69.61]	1.01 [91.09]	-0.07 [-3.35]
β_{SMB}	0.26 [10.33]	0.09 [4.44]	0.09 [3.95]	-0.05 [-2.69]	-0.12 [-7.62]	-0.38 [-11.91]
β_{HML}	0.11 [3.45]	0.21 [8.49]	0.09 [3.17]	-0.07 [-2.99]	-0.11 [-5.80]	-0.22 [-5.61]
β_{RMW}	-0.54 [-17.01]	-0.20 [-7.85]	-0.04 [-1.35]	0.16 [6.33]	0.22 [11.06]	0.76 [18.88]
β_{CMA}	-0.23 [-5.20]	-0.09 [-2.57]	0.11 [2.73]	0.04 [1.12]	0.07 [2.51]	0.30 [5.33]
β_{UMD}	-0.06 [-3.81]	-0.01 [-0.52]	-0.03 [-1.83]	-0.05 [-4.14]	0.03 [3.17]	0.09 [4.58]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1590	792	590	527	571	
me (\$10 ⁶)	1746	2174	2618	4507	5329	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ECR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.52 [3.20]	0.76 [5.09]	0.71 [6.09]	0.62 [5.33]	0.30 [3.27]	0.24 [2.66]
Quintile	NYSE	EW	0.41 [2.35]	0.50 [2.91]	0.37 [2.74]	0.26 [1.96]	-0.00 [-0.03]	-0.08 [-0.73]
Quintile	Name	VW	0.41 [2.16]	0.66 [3.77]	0.59 [4.18]	0.57 [3.93]	0.26 [2.04]	0.25 [1.93]
Quintile	Cap	VW	0.43 [2.93]	0.60 [4.30]	0.58 [4.95]	0.50 [4.24]	0.17 [1.83]	0.12 [1.29]
Decile	NYSE	VW	0.53 [2.77]	0.79 [4.40]	0.71 [4.70]	0.67 [4.41]	0.24 [1.90]	0.24 [1.84]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.49 [3.04]	0.74 [4.92]	0.68 [5.79]	0.62 [5.37]	0.34 [3.67]	0.29 [3.27]
Quintile	NYSE	EW	0.22 [1.27]	0.32 [1.81]	0.21 [1.45]	0.14 [1.03]		
Quintile	Name	VW	0.37 [1.99]	0.65 [3.63]	0.57 [3.96]	0.55 [3.81]	0.28 [2.16]	0.27 [2.08]
Quintile	Cap	VW	0.41 [2.80]	0.58 [4.07]	0.54 [4.58]	0.48 [4.17]	0.20 [2.19]	0.17 [1.82]
Decile	NYSE	VW	0.50 [2.60]	0.75 [4.14]	0.66 [4.37]	0.64 [4.21]	0.26 [2.04]	0.25 [1.99]

Table 3: Conditional sort on size and ECR

This table presents results for conditional double sorts on size and ECR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECR and short stocks with low ECR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ECR Quintiles					ECR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.21 [0.46]	0.45 [1.21]	0.64 [2.19]	0.87 [3.28]	1.04 [3.43]	0.84 [2.96]	1.11 [4.01]	0.89 [4.06]	0.85 [3.81]	0.40 [2.08]	0.38 [1.94]
	(2)	0.33 [0.83]	0.55 [1.81]	0.77 [2.91]	0.93 [3.57]	1.06 [3.74]	0.73 [3.28]	0.98 [4.58]	0.81 [4.83]	0.78 [4.61]	0.34 [2.58]	0.34 [2.50]
	(3)	0.33 [0.97]	0.66 [2.36]	0.75 [3.02]	0.96 [3.99]	1.03 [3.87]	0.70 [3.67]	0.89 [4.82]	0.74 [5.08]	0.69 [4.66]	0.30 [2.48]	0.27 [2.23]
	(4)	0.62 [2.01]	0.65 [2.63]	0.82 [3.64]	0.87 [3.78]	0.98 [4.03]	0.36 [2.36]	0.54 [3.68]	0.44 [3.51]	0.39 [3.06]	0.08 [0.76]	0.05 [0.48]
	(5)	0.43 [1.70]	0.60 [2.66]	0.68 [3.31]	0.90 [4.40]	0.90 [4.37]	0.47 [3.17]	0.63 [4.42]	0.65 [4.73]	0.54 [3.94]	0.21 [1.82]	0.14 [1.21]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ECR Quintiles					ECR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	438	452	456	458	450	39	46	54	62	61	
	(2)	132	133	133	133	133	88	91	94	97	96	
	(3)	90	90	90	90	90	152	158	161	164	165	
	(4)	74	74	75	75	74	325	331	346	347	351	
(5)	66	66	66	66	66	2075	1838	2300	3287	3645		

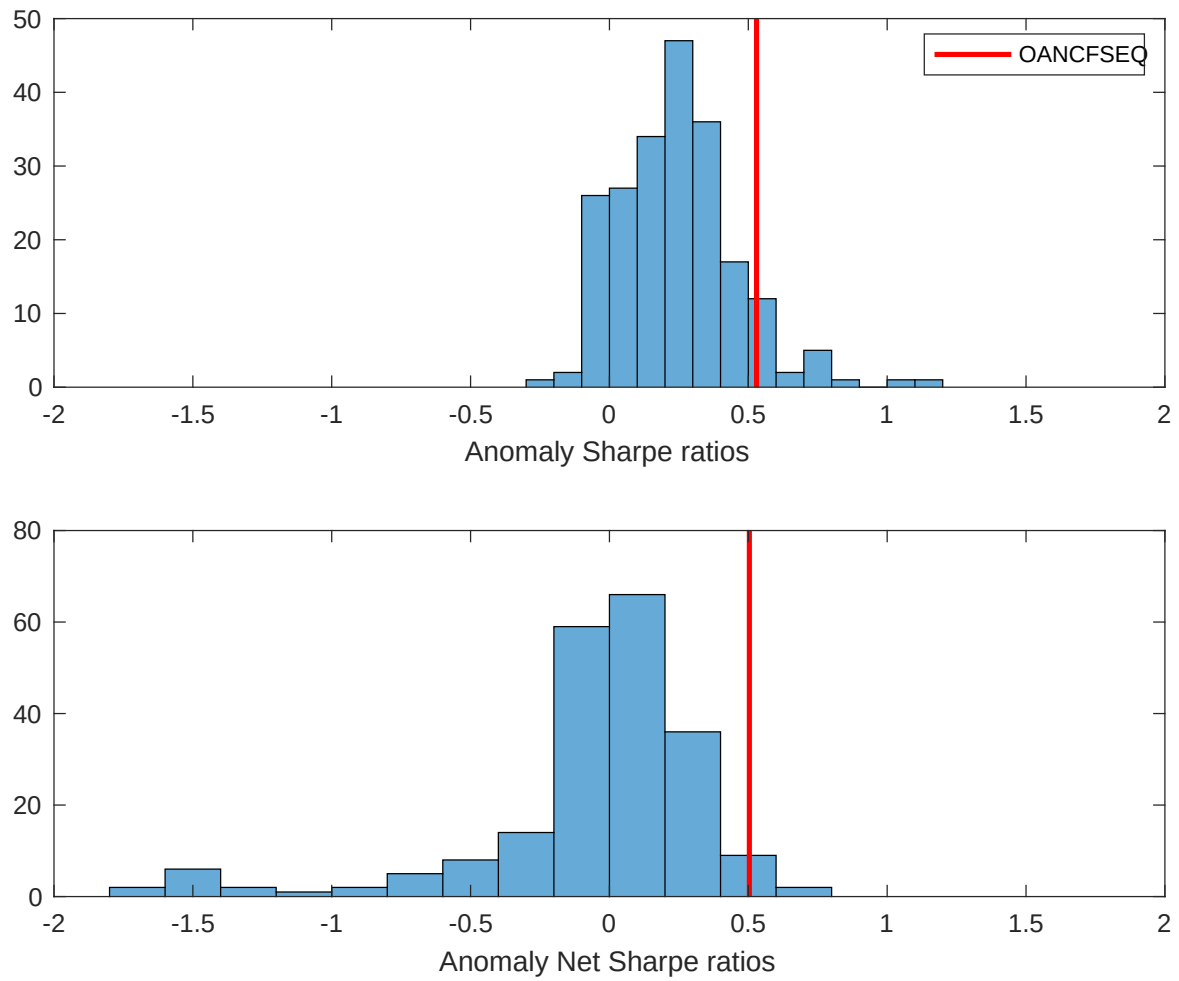


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

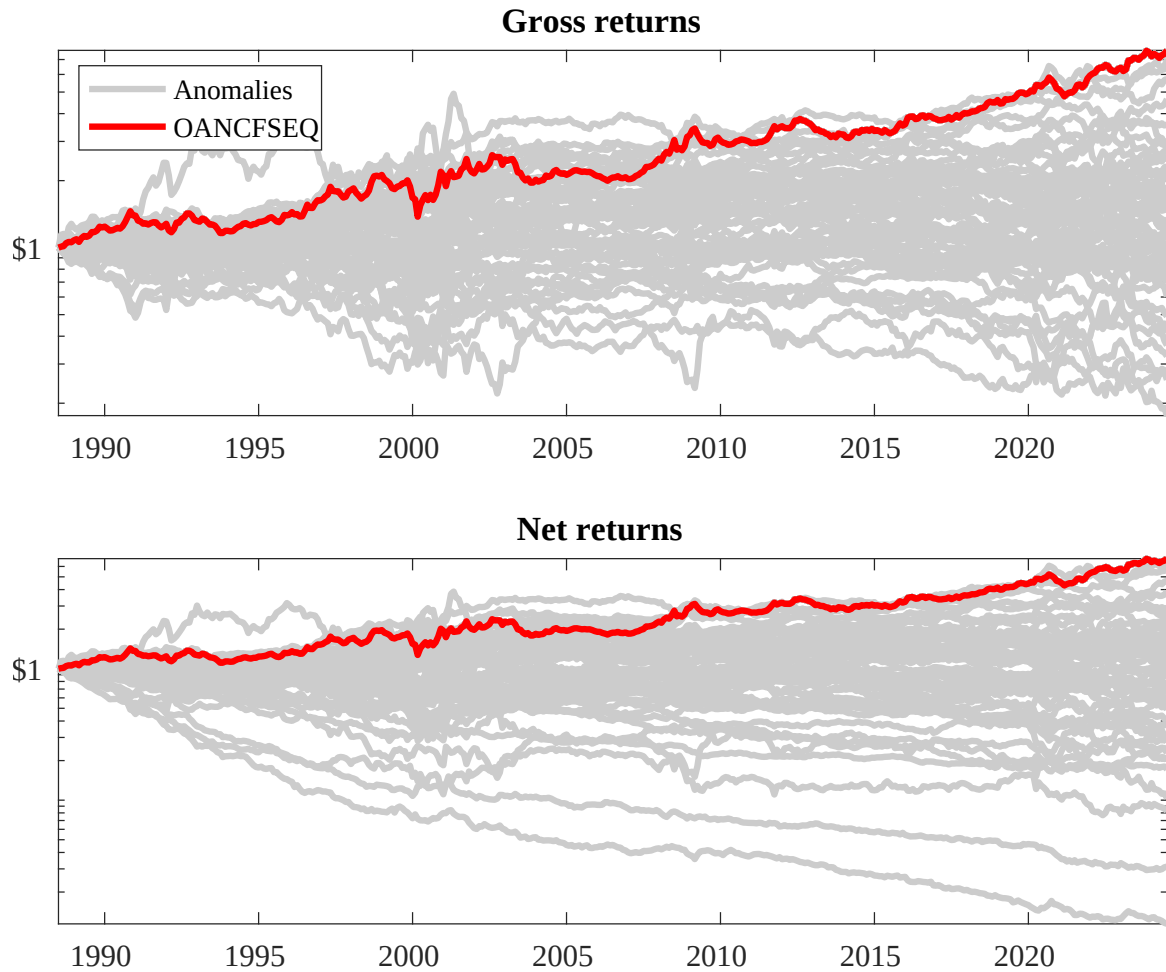
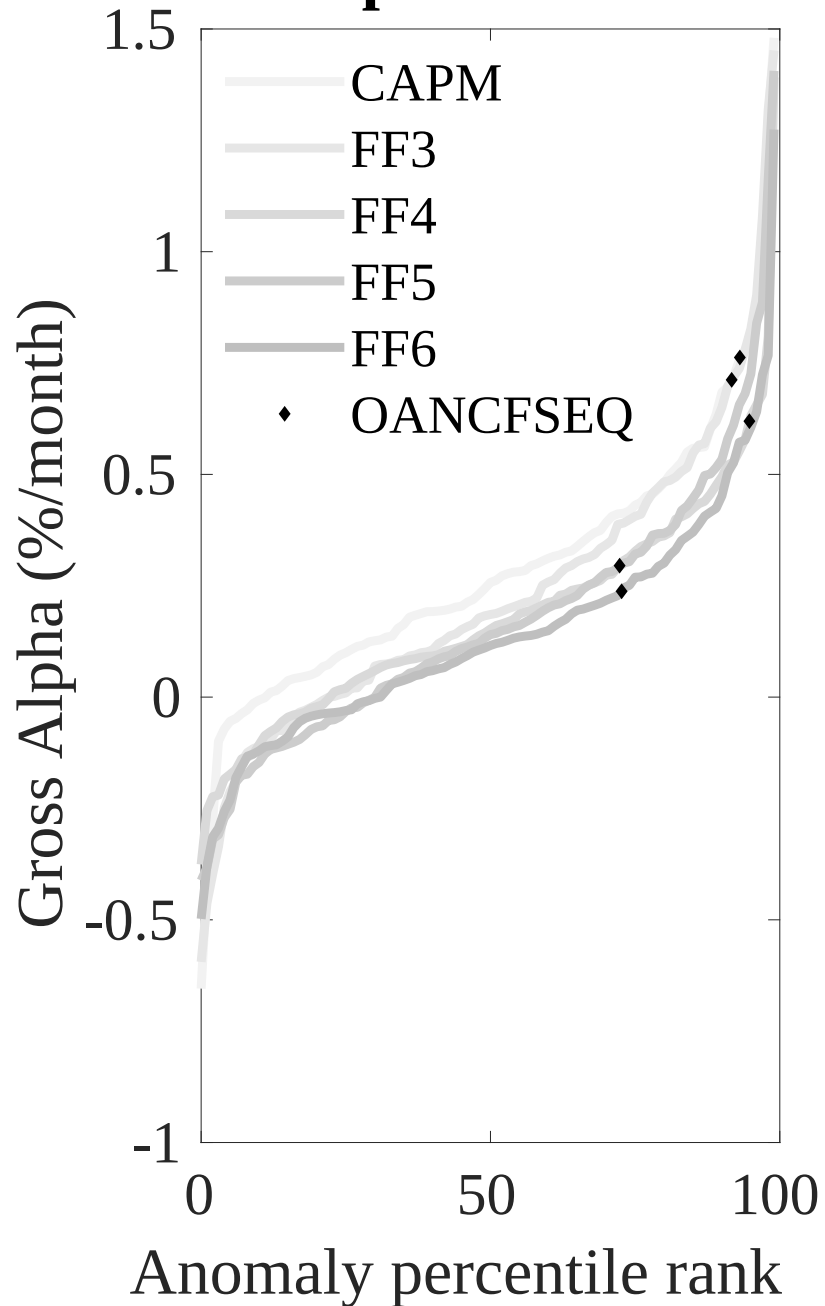


Figure 3: Dollar invested.

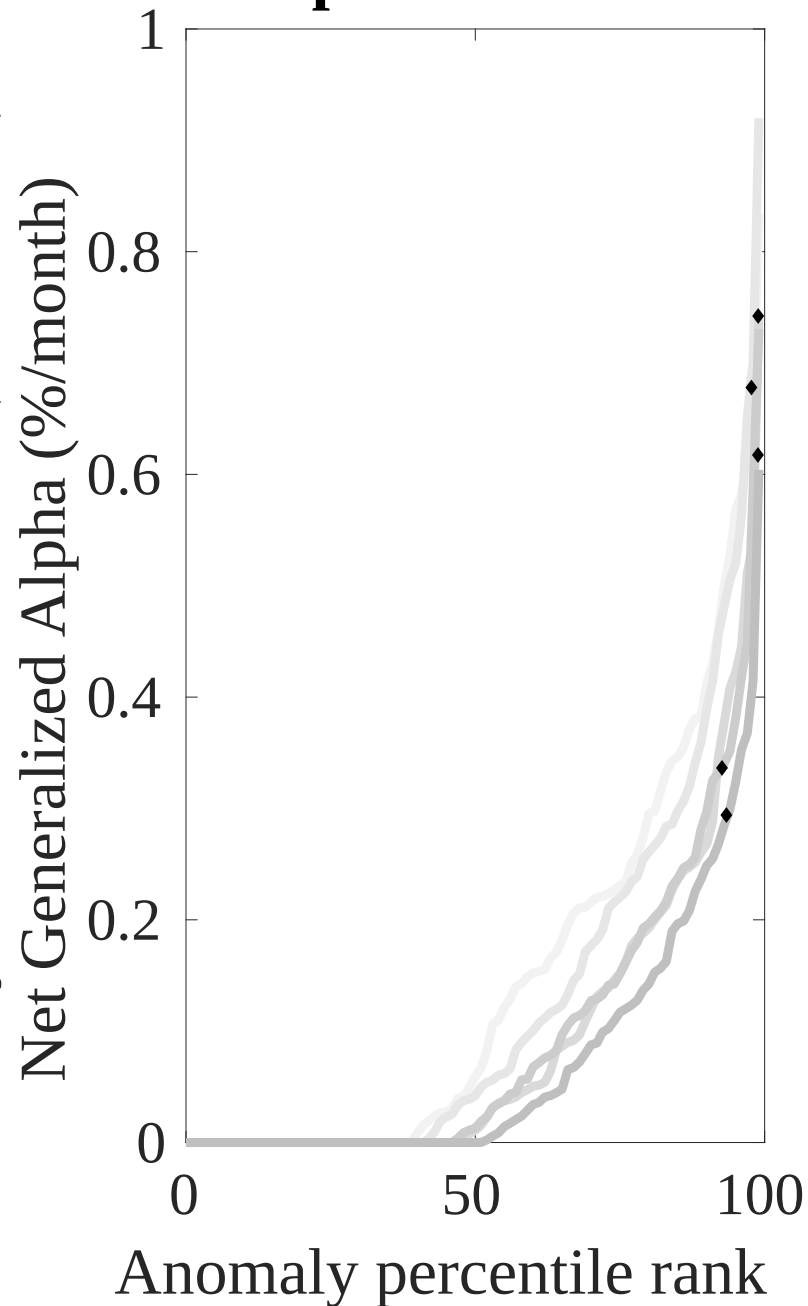
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



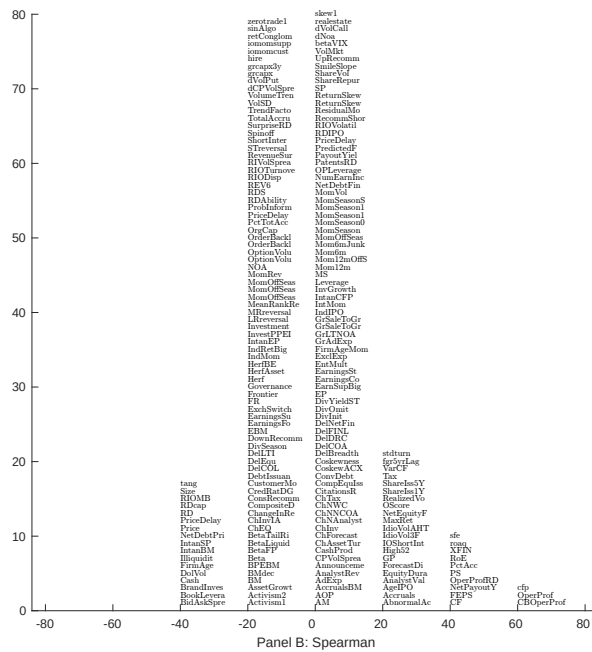
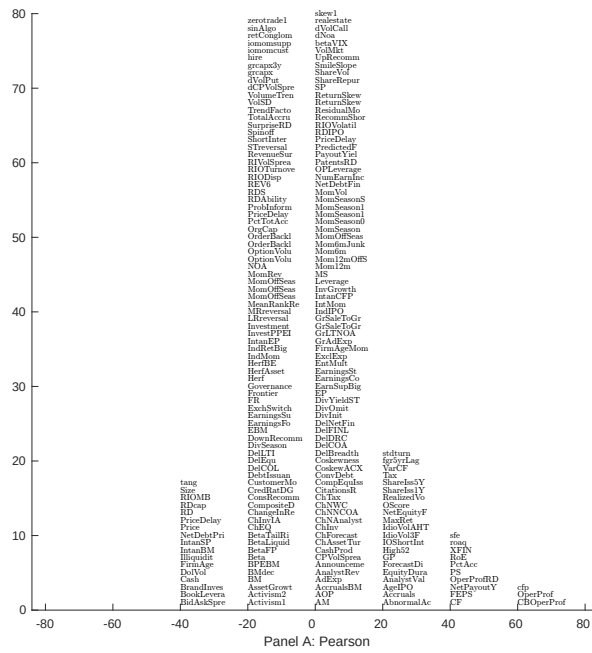


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with ECR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

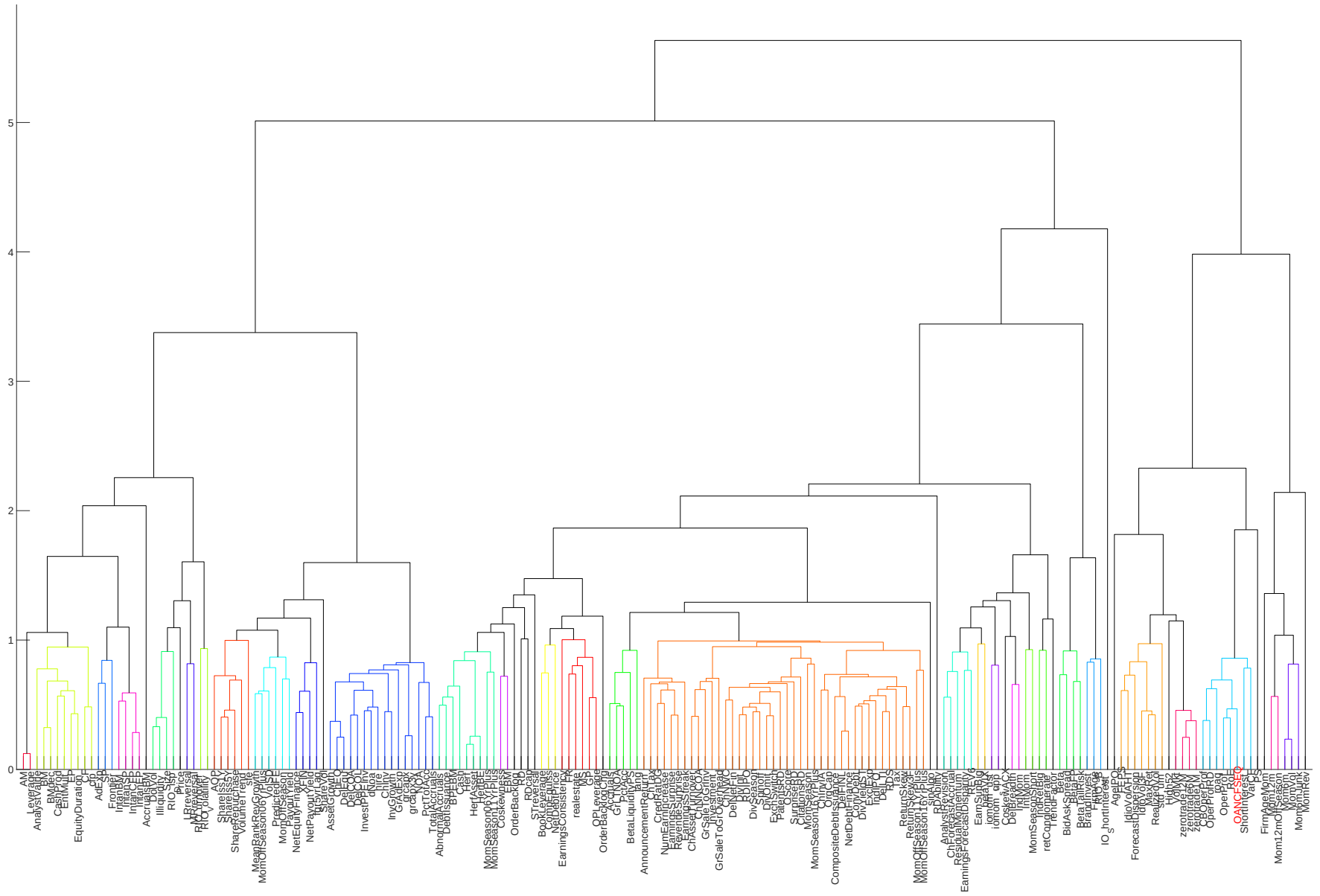


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

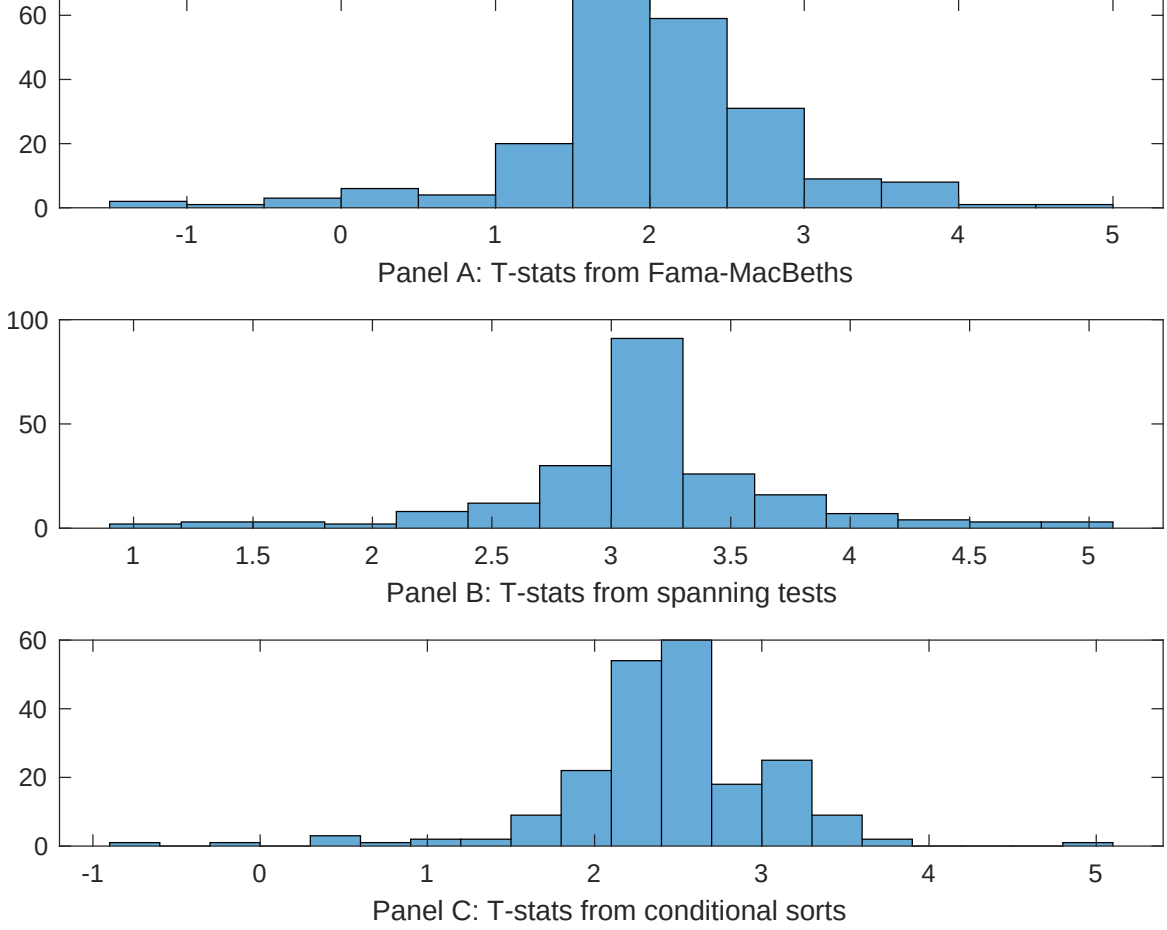


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ECR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECR} ECR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECR. Stocks are finally grouped into five ECR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ECR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ECR} ECR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.90 [3.02]	0.10 [3.47]	0.87 [2.53]	0.12 [4.25]	0.11 [4.73]	0.11 [3.99]	0.11 [5.10]
ECR	0.22 [2.56]	0.26 [2.49]	0.25 [3.22]	0.68 [0.07]	0.25 [3.88]	0.69 [0.99]	0.15 [2.16]
Anomaly 1	0.22 [2.96]						0.23 [0.32]
Anomaly 2		-0.84 [-0.16]					-0.13 [-2.35]
Anomaly 3			0.46 [0.69]				-0.17 [-0.60]
Anomaly 4				0.18 [6.02]			0.11 [4.36]
Anomaly 5					0.70 [1.17]		0.61 [0.76]
Anomaly 6						0.42 [2.80]	0.35 [3.09]
# months	427	432	427	432	427	427	427
$\bar{R}^2(\%)$	0	1	2	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ECR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ECR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015a\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.21 [2.49]	0.23 [2.57]	0.24 [2.68]	0.20 [2.24]	0.24 [2.68]	0.23 [2.53]	0.19 [2.28]
Anomaly 1	34.67 [8.75]						36.13 [7.70]
Anomaly 2		20.55 [3.83]					-5.03 [-0.84]
Anomaly 3			-2.34 [-0.68]				-17.57 [-4.43]
Anomaly 4				21.40 [4.98]			11.97 [2.57]
Anomaly 5					4.07 [1.41]		5.16 [1.61]
Anomaly 6						7.05 [1.77]	5.62 [1.40]
mkt	-3.39 [-1.61]	-4.37 [-1.88]	-7.72 [-3.26]	-4.11 [-1.81]	-5.51 [-2.17]	-6.19 [-2.69]	-3.15 [-1.35]
smb	-29.97 [-9.62]	-31.76 [-8.82]	-39.78 [-10.38]	-31.24 [-9.10]	-34.90 [-8.64]	-36.95 [-11.17]	-32.41 [-8.44]
hml	-16.44 [-4.50]	-19.61 [-5.03]	-20.96 [-5.29]	-19.10 [-4.96]	-22.41 [-5.61]	-19.00 [-4.60]	-12.21 [-3.09]
rmw	45.21 [8.81]	58.05 [9.44]	78.55 [14.14]	63.13 [13.41]	71.65 [14.12]	69.55 [12.80]	50.00 [7.83]
cma	23.06 [4.33]	33.27 [5.90]	31.59 [5.44]	17.19 [2.77]	29.34 [5.06]	30.46 [5.34]	17.54 [2.93]
umd	5.18 [2.73]	8.27 [4.18]	9.94 [4.68]	8.73 [4.50]	8.50 [4.06]	7.62 [3.42]	6.08 [2.86]
# months	428	432	428	432	428	428	428
$\bar{R}^2(\%)$	78	74	74	75	74	74	79

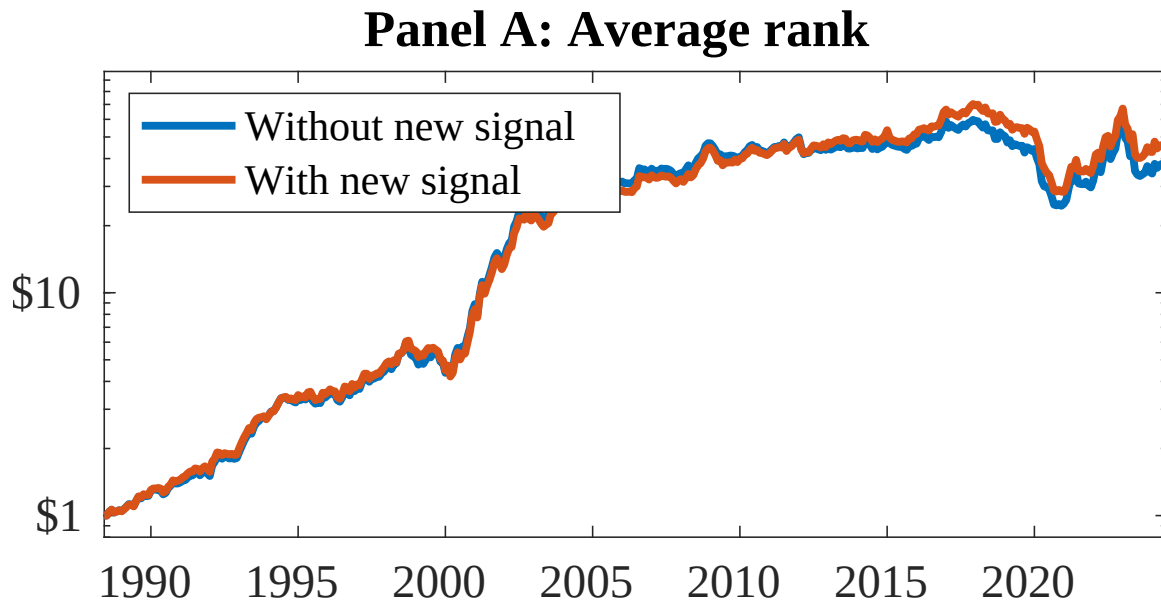


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ECR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Ball, R., Gerakos, J., Linnainmaa, J. T., and Nikolaev, V. V. (2015). Deflating profitability. *Journal of Financial Economics*, 117(2):225–248.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Daniel, K., Hirshleifer, D., and Subrahmanyam, A. (1998). Investor psychology and security market under- and overreactions. *Journal of Finance*, 53(6):1839–1885.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015a). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2015b). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

- Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.
- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H., Torous, W., and Valkanov, R. (2007). Do industries lead stock markets? *Journal of Financial Economics*, 83(2):367–396.
- Hou, K. and Moskowitz, T. J. (2006). Market frictions, price delay, and the cross-section of expected returns. *Review of Financial Studies*, 18(3):981–1020.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.
- Novy-Marx, R. and Velikov, M. (2016a). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2016b). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Richardson, S., Tuna, I., and Wysocki, P. (2010). Accounting anomalies and fundamental analysis: A review of recent research advances. *Journal of Accounting and Economics*, 50(2-3):410–454.
- Zhang, X. F. (2006). Information uncertainty and stock returns. *Journal of Finance*, 61(1):105–137.